Logic, Language and Philosophy Course

Extra Exercises

Meeting 29
December 29, 2025

1 Translation

Exercise 1 Translate the following sentences into the language of predicate logic. Define the domain and specify the translation key:

a. Everyone who admires everyone admires himself
b. Everyone who admires everyone else hates himself
c. There is only one uncaused cause

2 Model checking

Exercise 2 Consider the following model for a predicate logical language with a 1-place predicate letter A, and 2-place predicate letter R:

$$D = \{1, 2, 3, 4\}$$
$$I(A) = \{1, 4\}$$
$$I(R) = \{\langle 1,1\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 3,3\rangle, \langle 4,1\rangle, \langle 4,2\rangle, \langle 4,3\rangle, \langle 4,4\rangle\}$$

a. Draw a graph of the model
b. Which of the following formulas are true in the model? Motivate your answer.
   i. $\forall x(Ax \leftrightarrow \forall y Rxy)$
   ii. $\forall x(Ax \rightarrow \exists y \exists z(y \neq z \wedge Ryx \wedge Rzx))$
   iii. $\exists x \forall y(Ryy \leftrightarrow x \neq y)$

3 Validity and counterexamples

Exercise 3 Prove:

a. $\models \forall x Rxx \rightarrow \forall x \exists y Rxy$
b. $\exists x Ax \vee \exists x Bx, \forall x(Ax \leftrightarrow Bx) \models \exists x(Ax \wedge Bx)$
c. $\forall x(\phi \rightarrow \psi), [c/x]\phi \models [c/x]\psi$

Exercise 4 Prove by specifying a counterexample:

a. $\forall x(Ax \vee Bx) \not\models \forall x Ax \vee \forall x Bx$
b. $\forall x Rxx \not\models \forall x \forall y Rxy$
c. $\forall x \exists y(Rxy \wedge x \neq y), \forall x \forall y \forall z((Rxy \wedge Ryz) \rightarrow Rxz) \not\models \forall x \forall y(x = y \vee Rxy \vee Ryx)$

4 Relations and orderings

Exercise 5

a. Specify a domain on which the relation x *is at least as big as* y is anti-symmetric and one on which it is not.
b. Under which assumption(s) is the relation x *is related by blood to* y connected on the domain of people?

5 Natural deduction

Exercise 6 Make derivations in the system of natural deduction for the following argument schemes:

a. $\forall x \forall y Rxy \vdash \forall x Rxx$
b. $\exists x Ax \rightarrow \forall y By \vdash \forall x \forall y(Ax \rightarrow By)$
c. $\forall x \neg Rxx, Rab \vdash a \neq b$

6 Soundness and completeness

Exercise 7 Show that the following hold:

a. If $\vdash \forall x \phi$, then $\vdash \exists x \phi$
b. If $\vdash \phi$, then $\vdash \forall x([x/c]\phi)$, provided x is free for c in ϕ

Exercise 8 Show that the following hold:

a. If $\Gamma \models \phi \wedge \psi$, then $\Gamma \models \psi$
b. If $\Gamma \models \neg\neg\phi$, then $\Gamma \models \phi$

Exercise 9 Let Γ be a maximally consistent set, and ϕ and ψ arbitrary formulas. Show that the following holds:

a. $\neg\phi \in \Gamma$ iff $\phi \notin \Gamma$
b. $\phi \rightarrow \psi \in \Gamma$ iff $\phi \notin \Gamma$ or $\psi \in \Gamma$

7 Intensional logic

Exercise 10 Consider the following model for a modal propositional language with two atomic propositions p and q:

$$W = \{w, v, u\}$$
$$R = \{\langle w, w\rangle, \langle w, v\rangle, \langle w, u\rangle, \langle u, v\rangle, \langle v, u\rangle\}$$
$$V_w(p) = V_v(p) = V_v(q) = V_u(q) = 1; V_w(q) = V_u(p) = 0$$

1. Draw a graph of the model
b. Determine the truth values of:
   i. $\Box q$ and $\Diamond\Box q$ in w
   ii. $\Box\neg p$ and $\Box\Box\neg p$ in v and u
   iii. $\Diamond\Box q \rightarrow \neg q$ in w and u

Exercise 11 Show that the modal principle $\Box\phi \rightarrow \Box\Box\phi$ corresponds with transitivity of the accessibility relation R.

8 Philosophical papers

Exercise 12

i. How does Quine make use of Russell's theory of descriptions? Could he also use Frege's account of descriptions? Or Strawson's? Why (not)?
ii. Does Quine's criterion of ontological commitment tell us what there is? Is it supposed to?
iii. What does Tarski understand by a 'semantically closed language'?
iv. Russell's theory of descriptions is called 'contextual'. Explain. Is there a problem with compositionality? If so, what is it? If not, why not?
v. Is Strawson's approach to description semantic or pragmatic? What is Strawson view on the role of the principle of excluded middle?
vi. Engel views truth and validity as normative notions. Explain what he means with reference to the principle of bi-valence.

vii. Identify an argument in Frege's paper where he implicitly appeals to compositionality. What kind of compositionality is it?

viii. Can a rigid designator have descriptive content according to Kripke? If so, what kind of descriptive content is that?